Freight Rate Prediction: Validation Report

1. Data and split approach

The labeled data (`train-test.csv`) has 48,000 loads from 2025-01-01 to 2025-10-31. The loads to predict (`validation.csv`) are 12,000 loads from 2025-11-01 to 2025-12-31, so the final task is to predict **future** months from past data.

Because of this, I validated with a **time-based rolling split** instead of a random split. A random split would mix days from the same period into both train and test, which makes the scores look better than they would be on genuinely future data. For each of July, August, September and October, the model is trained only on the months before it and tested on that month. The reported numbers are the average of these four folds. The final model is then trained on all of January to October.

2. Data quality checks and handling

I checked the data for negative and zero values, missing values, duplicates and extreme rates (`src/explore.py`). Findings:
- **Negative weights:** 292 rows have negative weights, which is not realistic. I set them to missing and filled them with the median.
- **Missing values:** 300 weights and 374 market index values are missing. They are filled with the median.
- **Duplicates:** no duplicate rows or load IDs.
- **Extreme rates:** 335 loads (about 0.7%) have a rate per mile above $4.50, while the typical load is around $2.15. They are spread across months and I could not explain them with the available features, so I kept them and used an absolute-error loss that is not pulled strongly by outliers.

3. Model

I used scikit-learn's **HistGradientBoostingRegressor**, a gradient boosting model suited to tabular data with non-linear patterns. Settings: absolute-error loss, 200 iterations, learning rate 0.08, 31 leaves per tree, L2 regularization 1.0. The model predicts `log(1 + rate)`, which is converted back to dollars. Features are the coordinates, distance, weight, market index, quote signal, year, month and day of week, plus one-hot encoded pickup city, delivery city and equipment.

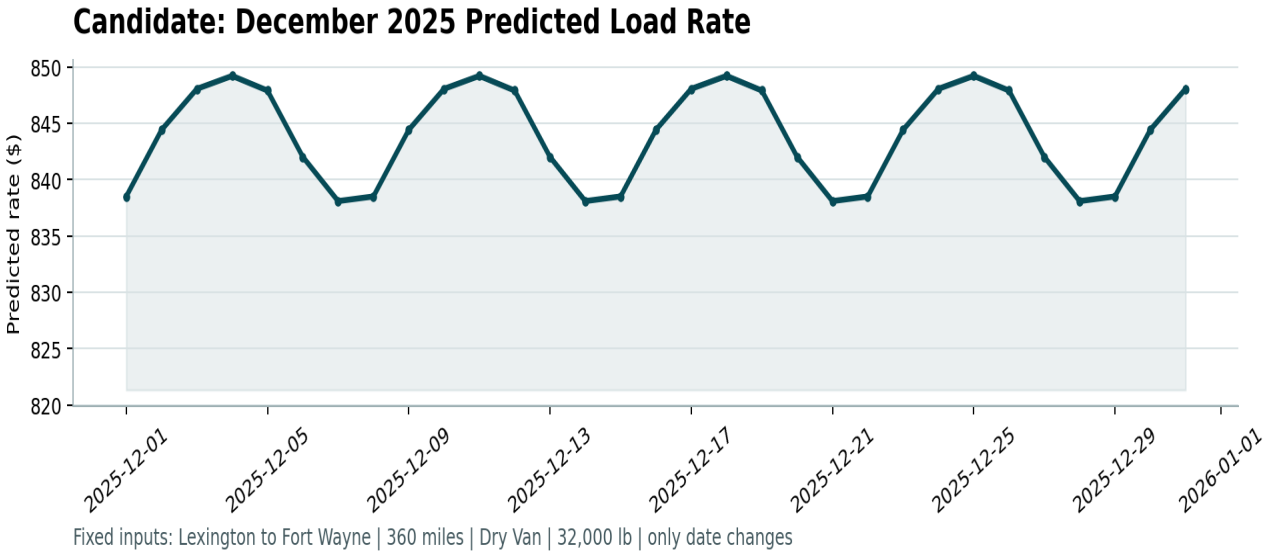
4. Validation results

Test month	Train rows	Test rows	MAE ($)	RMSE ($)	Median error	Mean error
2025-07	28,806	4,912	122.7	631.6	2.07%	4.72%
2025-08	33,718	4,759	120.7	617.5	2.59%	5.57%
2025-09	38,477	4,670	109.8	618.8	1.93%	4.38%
2025-10	43,147	4,853	108.5	648.0	1.78%	5.05%
Average			115.4	629.0	2.09%	4.93%

On average the model is off by about $115 per load (MAE), with a median error of 2.1%. RMSE is much higher (about $629) because it squares errors, so the small group of extreme-rate loads (about 0.7% of rows) dominates it. On normal loads (rate per mile of $4.50 or less) the RMSE is about $194.

5. December 2025 chart (produced by score.py)

The chart uses the fixed lane Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb, where only the date changes. The December inputs do not include market index or quote signal, so the December model uses only distance, weight, equipment, cities, and date features. Predicted rates range from $838 to $849 and follow a weekly pattern.



6. Limitations and next steps

- The November-December period is outside the training period, and the market index was lower there, so real errors may differ from the validation numbers.
- The extreme-rate loads limit how low RMSE can go with the current features.
- Possible improvements: add lane-level history features, use a daily market index (also for December), and test dropping `quote_signal`.